

CE 605: Applied Statistics

Project Report 2: Network Analysis

Network-Theoretic Analysis of Spatial Rainfall Correlation Structures Across India



Submitted by:

Abhijeet Kumar Gaurav

Roll No.: 24M0553

Under the Supervision of

Prof. [Bellie Sivakumar](#)

M.Tech in Transportation Systems Engineering

Department of Civil Engineering

Indian Institute of Technology Bombay

Mumbai – 400076, India

Autumn 2024

Abstract

Rainfall over India exhibits strong spatial correlation driven by shared climatic and geographic influences, yet conventional station-by-station statistical analysis struggles to capture the system-wide structure of these dependencies. This report applies complex network theory to monthly rainfall records from 20 meteorological grid stations (drawn from a national archive of over 4000 grids spanning January 1961 – December 2014) to represent stations as nodes and their pairwise rainfall correlations as edges. Adjacency structures are constructed at three correlation thresholds (0.2, 0.3, and 0.5), and the resulting networks are characterised using degree centrality, clustering coefficient, average shortest path length, global efficiency, and the harmonic mean of shortest path lengths. Results show that low thresholds produce a densely connected, near-complete network in which nearly all stations are reachable within a single hop, while progressively higher thresholds isolate a smaller set of strongly correlated “core” stations and fragment the remainder into sparser clusters. These findings illustrate how threshold selection governs the trade-off between network-wide connectivity and the specificity of identified relationships, offering a scalable, data-driven framework for studying regional rainfall dependence relevant to water-resource planning and climate-risk assessment.

Keywords: Complex Network Theory; Rainfall Correlation; Degree Centrality; Clustering Coefficient; Global Efficiency; Spatial Statistics

Contents

1	Introduction	1
2	Study Area and Data Collection	1
2.1	Study Area	1
2.2	Data Collection	2
3	Methodology	2
3.1	Network Theory Overview	2
3.2	Key Network Metrics	2
3.2.1	Degree Centrality (DC)	2
3.2.2	Clustering Coefficient (CC)	3
3.2.3	Shortest Path Length	3
3.2.4	Global Efficiency	3
3.2.5	Harmonic Mean of Shortest Path Lengths	4
3.3	Steps in Analysis	4
4	Results	4
4.1	Correlation Matrix and Network Formation	4
4.2	Global Efficiency and Harmonic Mean	5
4.3	Degree Centrality and Clustering Coefficient	6
4.4	Shortest Path Length and Network Visualisation	7
5	Discussion	9
5.1	Summary of Findings	11
6	Conclusions	12

1 Introduction

This project examines the spatial structure of monthly rainfall variability across India through the lens of complex network theory. Rather than treating each meteorological station in isolation, the analysis represents each station’s rainfall record as a *node* and quantifies the statistical similarity between every pair of stations as a weighted or thresholded *edge*. This graph-based representation converts a purely statistical question, namely “how correlated is rainfall across space?”, into a structural one that can be probed with the well-established toolkit of network science.

Framing rainfall data as a network is useful for several reasons:

1. **Modelling interconnectedness.** Representing each grid as a node and each significant correlation as an edge makes it possible to visualise which regions move together and which remain comparatively isolated, revealing how weather patterns propagate spatially.
2. **Identifying patterns and anomalies.** Metrics such as degree centrality and the clustering coefficient surface structure, for instance groups of stations sharing a climatic driver, that is difficult to see in a raw correlation table.
3. **Assessing vulnerability and resilience.** Highly connected (high-centrality) stations represent critical nodes for regional water-resource management; understanding which stations play this role informs planning for droughts and floods.
4. **Supporting real-time monitoring.** Because the network can, in principle, be updated as new data arrives, the same framework supports continuous tracking of how correlation structure evolves, which is valuable for early warning of shifts in rainfall distribution.
5. **Feeding integrated climate models.** Quantified spatial dependence structures can be incorporated into broader climate models that combine rainfall with other variables such as temperature and humidity.

The remainder of this report is organised as follows: Section 2 describes the study area and data; Section 3 presents the network-theoretic methodology and the metrics used; Section 4 reports the results across three correlation thresholds; Section 5 discusses the implications of threshold choice; and Section 6 concludes.

2 Study Area and Data Collection

2.1 Study Area

The study area covers India, where monthly rainfall has been digitised for over **4,000 meteorological grids**. Each grid represents a specific geographic location and captures a localised rainfall time series.

2.2 Data Collection

The parent dataset spans **54 years**, from **January 1961 to December 2014**, giving 648 monthly records per grid. For this project, a subset of **20 rainfall stations** was assigned, allowing a focused, computationally tractable analysis while preserving the spatial diversity of the national dataset.

3 Methodology

This analysis uses network theory to explore spatial relationships within rainfall data across different regions of India. Network theory models real-world systems as a collection of nodes and edges, enabling relationships, dependencies, and structural properties of the system to be captured quantitatively.

3.1 Network Theory Overview

Each node in the constructed network represents a rainfall station (grid point), and each edge indicates a statistically significant correlation between the rainfall records of two stations. Strong correlations suggest similar rainfall behaviour, which may be driven by shared climatic influences or geographic proximity. Figure 1 shows a simple illustrative network with six nodes and eight links, of the same type constructed in this study at a larger scale.

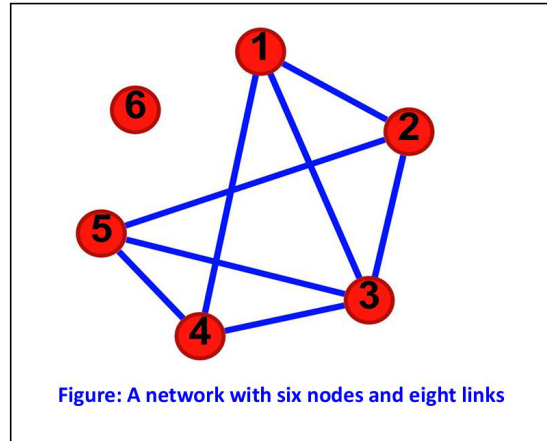


Figure 1: Illustrative network diagram showing nodes and links.

3.2 Key Network Metrics

3.2.1 Degree Centrality (DC)

Degree centrality measures the number of direct connections a node has with other nodes in the network. In the rainfall network, it indicates how many other stations a given station is closely

related to, based on a chosen correlation threshold.

$$DC = \frac{k}{N - 1}$$

where k is the number of neighbours of node i and N is the total number of nodes in the network. A higher degree centrality indicates that a station's rainfall pattern is shared with more locations in the network.

Procedure: (i) compute the pairwise correlation between all station pairs; (ii) apply a threshold (0.2, 0.3, or 0.5) to decide which correlations are considered significant; (iii) build the binary adjacency matrix; (iv) count, for each node, the number of connections above the threshold and apply the formula above.

3.2.2 Clustering Coefficient (CC)

The clustering coefficient quantifies the tendency of a node's neighbours to form connections among themselves, i.e. to form local clusters. A higher clustering coefficient indicates that a station's correlated neighbours are also correlated with each other, suggesting a localised region of similar rainfall behaviour.

$$c_i = \frac{2E_i}{k_i(k_i - 1)}$$

where E_i is the number of edges that actually exist among node i 's k_i neighbours.

3.2.3 Shortest Path Length

The shortest path length between two nodes is the minimum number of edges that must be traversed to connect them. In this context, it indicates how directly rainfall influence or similarity may propagate from one station to another; shorter paths suggest closer correlation or regional influence.

The **average shortest path length** L measures the overall efficiency of the network in terms of connectivity:

$$L = \frac{1}{N(N - 1)} \sum_{i \neq j} l_{ij}$$

where l_{ij} is the shortest path length between nodes i and j . A smaller L implies stations are closely interconnected on average, while a larger L suggests stations are relatively isolated from one another in terms of rainfall similarity.

3.2.4 Global Efficiency

Global efficiency measures how efficiently influence can spread across the network by averaging the *reciprocal* of shortest path lengths over all node pairs:

$$E_{glob} = \frac{1}{N(N - 1)} \sum_{i \neq j} \frac{1}{l_{ij}}$$

Higher global efficiency indicates that stations are well connected even across long (graph-distance) separations, facilitating rapid transmission of climatic influence across the network.

3.2.5 Harmonic Mean of Shortest Path Lengths

The harmonic mean of shortest path lengths is a robust connectivity measure that reduces the influence of outliers or disconnected node pairs: $HM = 1/E_{glob}$. Lower harmonic-mean values indicate that nodes can generally reach one another more easily, suggesting a well-integrated network.

3.3 Steps in Analysis

The overall analysis pipeline proceeds through four stages: **data preparation** → **network formation** → **computation of network metrics** → **results visualisation**. Correlation thresholds of **0.2**, **0.3**, and **0.5** were applied at the network-formation stage to study how connectivity structure changes as the significance criterion is tightened.

4 Results

After preprocessing and cleaning, a refined dataset of valid monthly rainfall records from 20 grids across India (January 1961 – December 2014) was obtained, with each grid represented as a node in the correlation network.

4.1 Correlation Matrix and Network Formation

Figure 2 shows the pairwise Pearson correlation coefficients between all 20 stations as a heatmap, using a diverging (“coolwarm”) colour scale: warmer colours indicate positive correlation and cooler colours indicate negative correlation. Diagonal cells are left blank, since a station’s self-correlation is trivially 1 and is not relevant to the inter-station analysis.

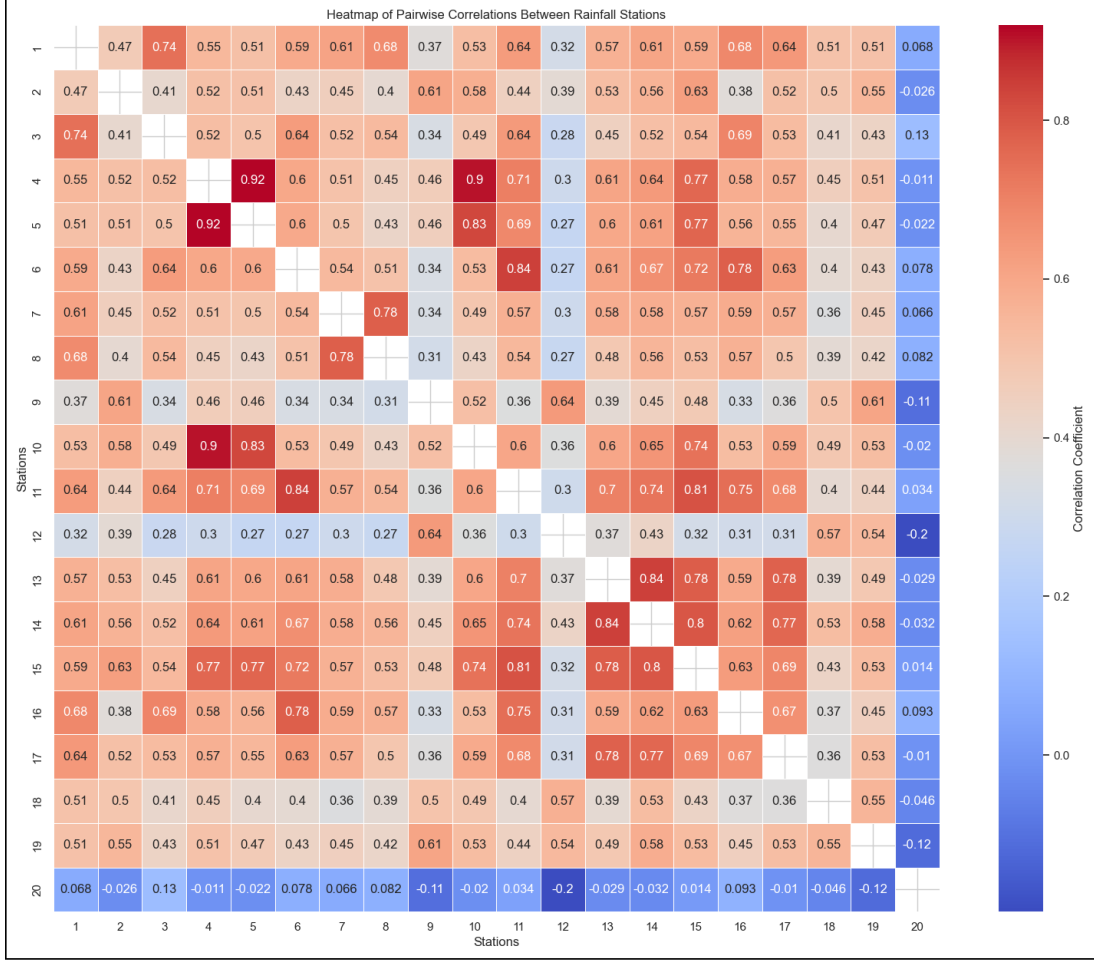


Figure 2: Heatmap of pairwise correlations between the 20 rainfall stations.

The adjacency matrix at each threshold is derived directly from this correlation matrix: an edge is placed between two stations whenever their correlation coefficient exceeds the chosen threshold. As expected, higher thresholds progressively remove weaker edges, isolating nodes that fail to meet the stricter criterion, while lower thresholds retain a much denser set of connections.

4.2 Global Efficiency and Harmonic Mean

Table 1 summarises the global efficiency and harmonic mean of the network at each threshold.

Table 1: Global efficiency and harmonic mean across correlation thresholds.

Threshold	Global Efficiency	Harmonic Mean
0.2	0.450000	2.222222
0.3	0.442105	2.261905
0.5	0.360965	2.770352

As the threshold increases, global efficiency decreases and the harmonic mean of shortest path lengths increases, reflecting the network's progressive fragmentation as only the strongest

correlations are retained.

4.3 Degree Centrality and Clustering Coefficient

Table 2 reports the degree centrality (DC) and clustering coefficient (CC) of each node at thresholds 0.2, 0.3, and 0.5. At threshold 0.2, nodes 1–19 are near-fully connected ($DC \approx 0.947$) and node 20 is isolated ($DC = 0$), giving a network-average DC of 0.900; at threshold 0.3 the network-average DC is 0.868; and at threshold 0.5 it falls to 0.563.

Table 2: Degree centrality (DC) and clustering coefficient (CC) by node across the three correlation thresholds.

Node	Threshold 0.2		Threshold 0.3		Threshold 0.5	
	DC	CC	DC	CC	DC	CC
1	0.947368	1.000000	0.947368	0.960784	0.789474	0.733333
2	0.947368	1.000000	0.947368	0.960784	0.473684	0.777778
3	0.947368	1.000000	0.894737	1.000000	0.578947	0.945455
4	0.947368	1.000000	0.894737	1.000000	0.736842	0.824176
5	0.947368	1.000000	0.894737	1.000000	0.684211	0.884615
6	0.947368	1.000000	0.894737	1.000000	0.684211	0.897436
7	0.947368	1.000000	0.894737	1.000000	0.631579	0.924242
8	0.947368	1.000000	0.894737	1.000000	0.421053	1.000000
9	0.947368	1.000000	0.947368	0.960784	0.263158	0.600000
10	0.947368	1.000000	0.947368	0.960784	0.684211	0.756410
11	0.947368	1.000000	0.947368	0.960784	0.684211	0.897436
12	0.947368	1.000000	0.631579	1.000000	0.157895	1.000000
13	0.947368	1.000000	0.947368	0.960784	0.631579	0.909091
14	0.947368	1.000000	0.947368	0.960784	0.842105	0.700000
15	0.947368	1.000000	0.947368	0.960784	0.789474	0.780952
16	0.947368	1.000000	0.947368	0.960784	0.684211	0.897436
17	0.947368	1.000000	0.947368	0.960784	0.736842	0.824176
18	0.947368	1.000000	0.947368	0.960784	0.263158	0.600000
19	0.947368	1.000000	0.947368	0.960784	0.526316	0.600000
20	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000

At threshold 0.2, nearly every station shares a strong correlation with almost every other station, so degree centrality is nearly uniform and clustering coefficients saturate at 1.0 (excluding the isolated node 20). As the threshold is raised to 0.3 and then 0.5, degree centrality values spread out considerably. Figure 3 compares degree centrality node-by-node across all three thresholds, revealing a smaller set of consistently well-connected “hub” stations (e.g. nodes 1, 14, and 15 remain high at threshold 0.5) alongside stations that become comparatively peripheral.

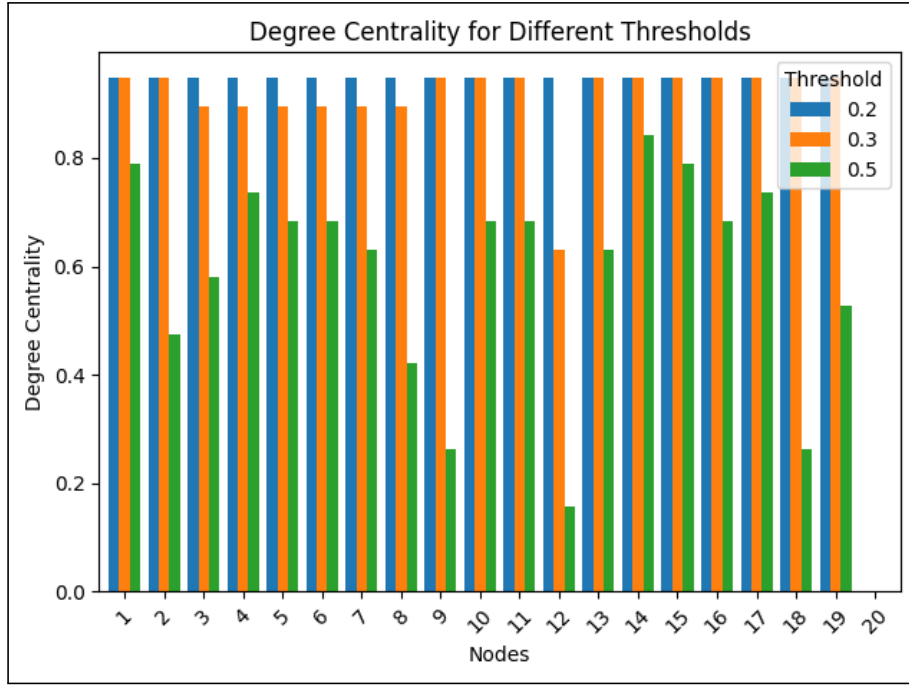


Figure 3: Degree centrality by node across the three correlation thresholds.

4.4 Shortest Path Length and Network Visualisation

Figures 4–6 visualise the resulting networks at each threshold. At threshold 0.2 the network is near-complete, with almost every pair of stations connected directly (a shortest path length of 1). As the threshold rises to 0.3 and 0.5, edges are progressively pruned, some shortest paths lengthen to 2 or 3 hops, and the network visibly sparsifies, with node 20 remaining disconnected across all thresholds.

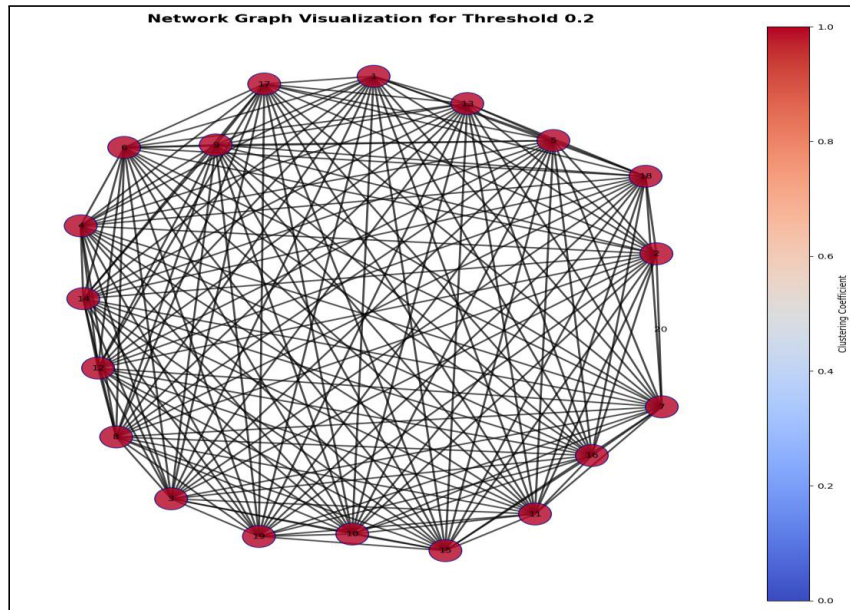


Figure 4: Network graph at threshold 0.2: a dense, near-complete graph.

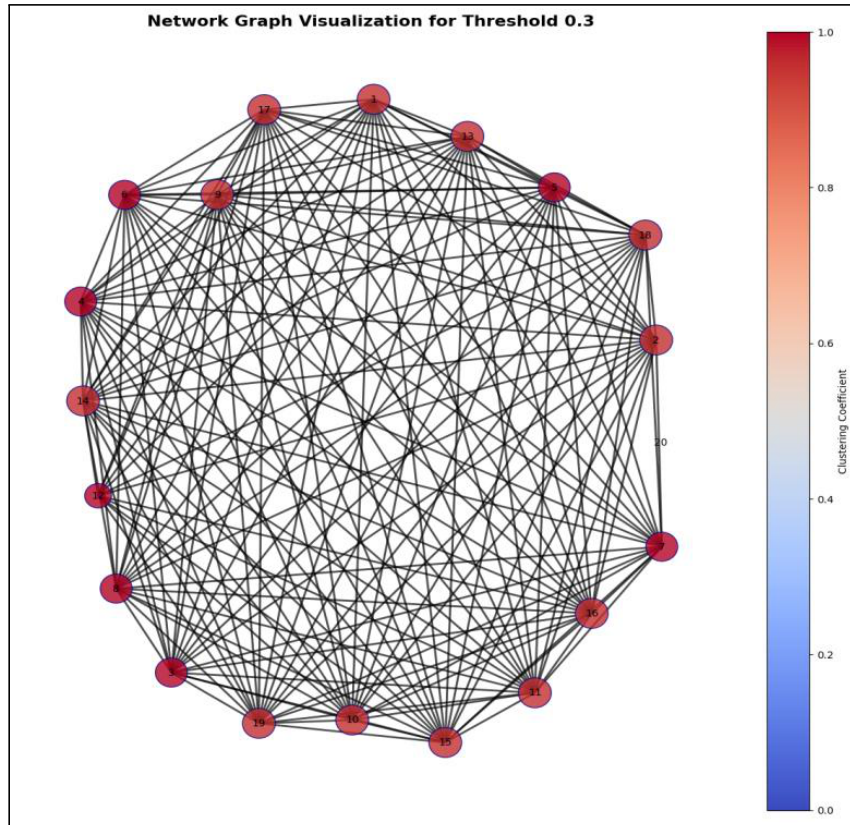


Figure 5: Network graph at threshold 0.3: moderately dense, with a small number of edges pruned.

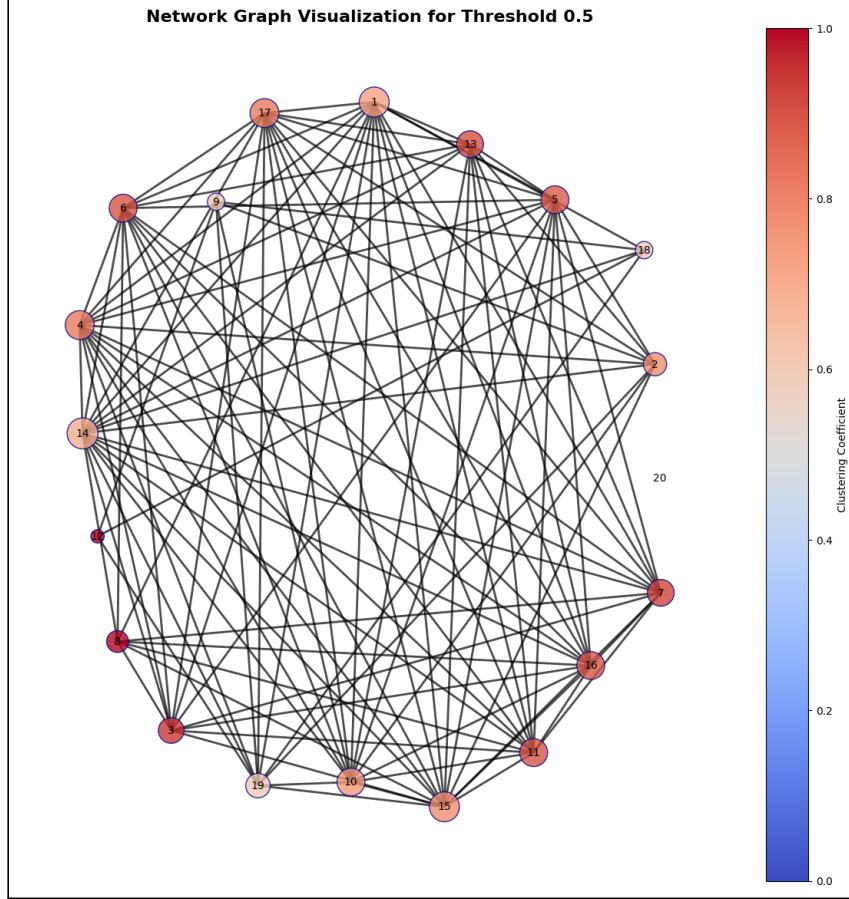


Figure 6: Network graph at threshold 0.5: a sparser, more selective network retaining only strongly correlated station pairs.

5 Discussion

Adjacency structure. As the threshold value increases, the adjacency matrix shows progressively decreased node connectivity. At threshold 0.2, nearly all stations are interconnected, reflecting even weak correlations; at 0.3, some connections begin to drop, reducing network density; at 0.5, only the strongest correlations remain, producing a sparser, more selective network. The choice of threshold therefore materially impacts network structure: lower thresholds capture broad interconnectivity, while higher thresholds focus on the most significant relationships, making the network more robust and manageable for in-depth, targeted analysis.

Global efficiency. Global efficiency decreases as the threshold increases. At 0.2 the network exhibits high global efficiency, meaning nodes can reach one another relatively quickly; at 0.5, efficiency is reduced as fewer paths are available. Lower thresholds thus provide a more connected network with better node accessibility, while higher thresholds, though reducing global efficiency, may capture only the most reliable relationships, a balance that must be struck depending on the analysis objective.

Harmonic mean. The harmonic mean of global efficiency is highest at threshold 0.2 and decreases with increasing thresholds, reflecting the higher connectivity available at lower thresholds

and the resulting shorter, more direct paths between stations. High harmonic-mean values at low thresholds indicate robust interconnectedness, valuable for analyses in which faster propagation of influence across the network is important.

Degree centrality. Degree centrality decreases as thresholds increase. At 0.2, central nodes have the highest degree centrality and serve as primary hubs; at 0.3 and 0.5, fewer nodes maintain high centrality, emphasising only the most critical connections. This shows which nodes are most influential within the network: lower thresholds highlight broad connections, while higher thresholds reveal core nodes with significant, and presumably more climatically meaningful, connections.

Clustering coefficient. The clustering coefficient is highest at lower thresholds, particularly 0.2, indicating that local clusters of interconnected nodes are prevalent. As the threshold increases, clustering decreases, showing more isolated nodes and fewer clusters. Higher clustering coefficients at low thresholds suggest denser local groups, useful for analysing community-based rainfall patterns; higher thresholds break these clusters apart, isolating key regions.

Shortest path length. Shortest path lengths increase with higher thresholds. At 0.2, nodes are generally close to each other, enabling rapid interaction across the network; at 0.5, the network's average path length grows, with some nodes becoming more isolated or reachable only through extended paths. Shorter paths at low thresholds indicate easy access and interconnectedness across the network, while higher thresholds create distinct clusters reflecting regions with particular rainfall patterns and more limited broader connectivity.

5.1 Summary of Findings

Table 3: Summary of network metrics across correlation thresholds.

Metric		Threshold 0.2	Threshold 0.3	Threshold 0.5	Remark
Adjacency Matrix		High connectivity, dense network	Moderate connectivity, less dense	Sparse network, limited connections	Lower thresholds capture even weak correlations that may reflect broad seasonal or regional patterns; higher thresholds isolate closely linked stations likely sharing climatic or topographic drivers.
Global Efficiency	Efficiency	High efficiency, easy reachability	Moderate efficiency	Low efficiency, selective paths	High efficiency at low thresholds implies rainfall patterns are interconnected across many stations; efficiency drops at high thresholds as only tightly correlated stations remain linked.
Harmonic Mean		High, robust interconnectedness	Moderate	Low	Lower thresholds reveal a highly interconnected network where changes at one station may broadly impact others; higher thresholds filter out weaker connections.
Degree Centrality		High centrality for several nodes	Moderate centrality	Few high-centrality nodes	Stations with high centrality at low thresholds act as rainfall-pattern hubs representing broad regional trends; only a few central stations remain connected at higher thresholds.
Clustering Coefficient		High clustering, local clusters present	Moderate clustering	Low clustering, isolated groups	High clustering at low thresholds implies groups of stations with similar rainfall patterns, possibly reflecting shared microclimates; higher thresholds show more distinct, localised clusters.
Shortest Path Length		Short paths, close accessibility	Moderate path lengths	Longer paths, limited accessibility	Shorter paths at low thresholds suggest many stations share similar rainfall trends; some stations become isolated at higher thresholds, representing distinct or unique rainfall regimes.

6 Conclusions

Using varying correlation thresholds, this network analysis of 20 rainfall stations provides a nuanced, structural view of rainfall correlation across regions of India. The network becomes progressively more selective at intermediate and high thresholds (0.3 and 0.5), retaining only the most strongly correlated stations and highlighting core groups with tightly coupled rainfall patterns, likely driven by shared geographic or topographic factors such as proximity or similar altitude. These clusters reflect microclimatic influences relevant to localised rainfall trend studies, water-resource management, flood forecasting, and climate-adaptation planning.

The variation in global efficiency and harmonic mean across thresholds highlights the fundamental trade-off between network-wide connectivity and specificity. Lower thresholds promote network-wide efficiency and interconnectedness, useful for understanding broad, interconnected rainfall dynamics, while higher thresholds reveal critical hubs and key stations with strong, direct correlations, providing a focused view of specific, highly correlated rainfall patterns. This selective structure at higher thresholds is valuable for identifying regions with distinct or isolated rainfall behaviour that may be susceptible to unique climate risks.

Overall, this analysis underscores the importance of selecting an appropriate correlation threshold based on the specific study objective: lower thresholds are preferable for capturing comprehensive regional patterns, while higher thresholds enable focused, in-depth analysis of localised rainfall phenomena. By applying network theory to rainfall data, this project demonstrates a scalable framework for analysing spatial dependencies that can support climate research and inform regional water-policy and planning decisions.

References

- [1] Chen, G., Wang, X., & Li, X. *Fundamentals of Complex Networks: Models, Structures and Dynamics*.
- [2] Boccaletti, S., Latora, V., Moreno, Y., Chavez, M., & Hwang, D. U. (2006). Complex networks: Structure and dynamics. *Physics Reports*, 424(4–5), 175–308.
- [3] Newman, M. E. J. (2010). *Networks: An Introduction*. Oxford University Press.
- [4] Fang, K., Ma, J., Yang, J., & Zhao, Q. (2015). Interannual variability and climate controls on streamflow and flooding in China’s paddy field areas. *Journal of Hydrology*, 521, 275–286.
- [5] Graphical representation of genetic code algebra (2023). <https://core.ac.uk/download/579840204.pdf>